MYSQL_PRACTICAL_Q01

```
--------------
CREATE TABLE countries (
    country_id VARCHAR(2),
    country_name VARCHAR(40),
    region_id DECIMAL(10,0)
)
--------------

--------------
DESCRIBE `countries`
--------------

+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| country_id   | varchar(2)    | YES  |     | NULL    |       |
| country_name | varchar(40)   | YES  |     | NULL    |       |
| region_id    | decimal(10,0) | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+

D:\mysql>
```

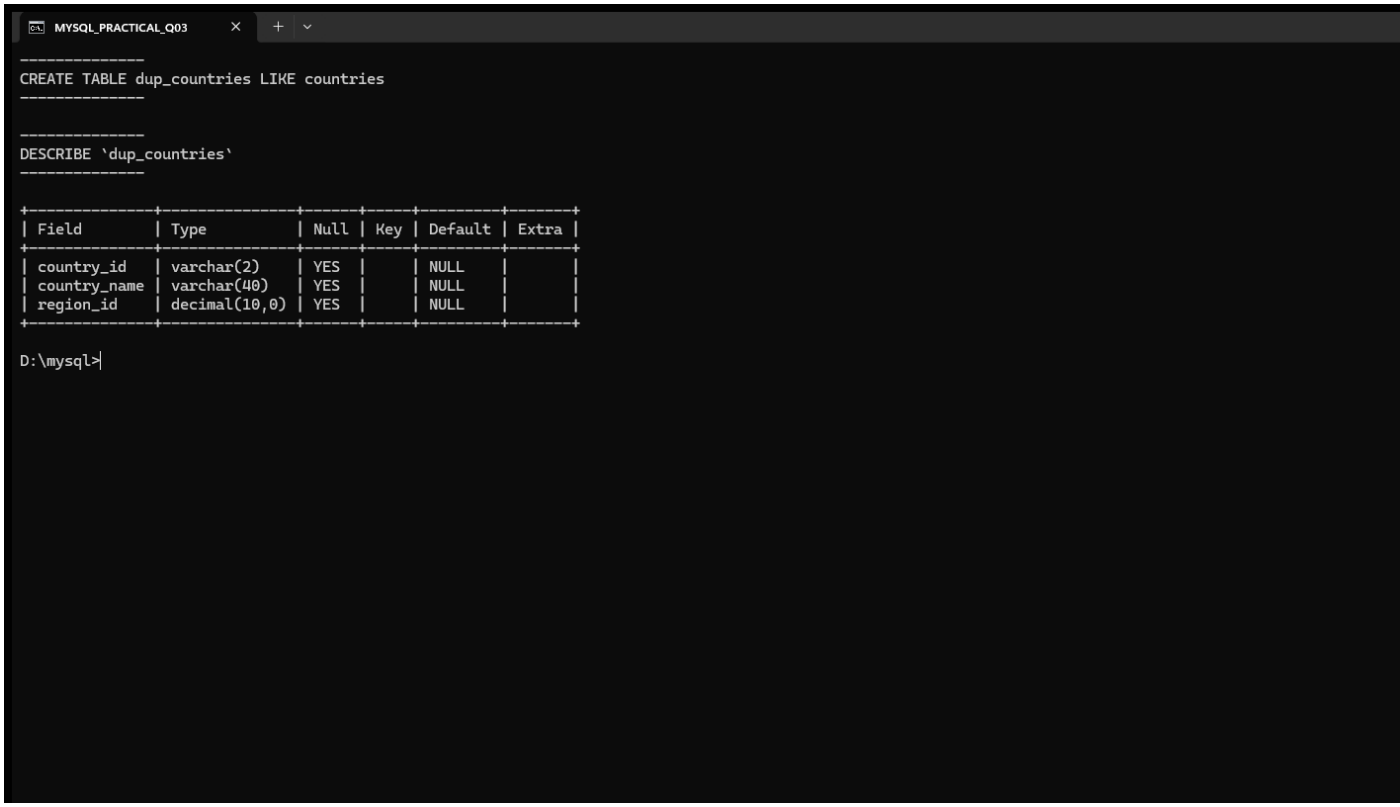
MYSQL_PRACTICAL_Q02

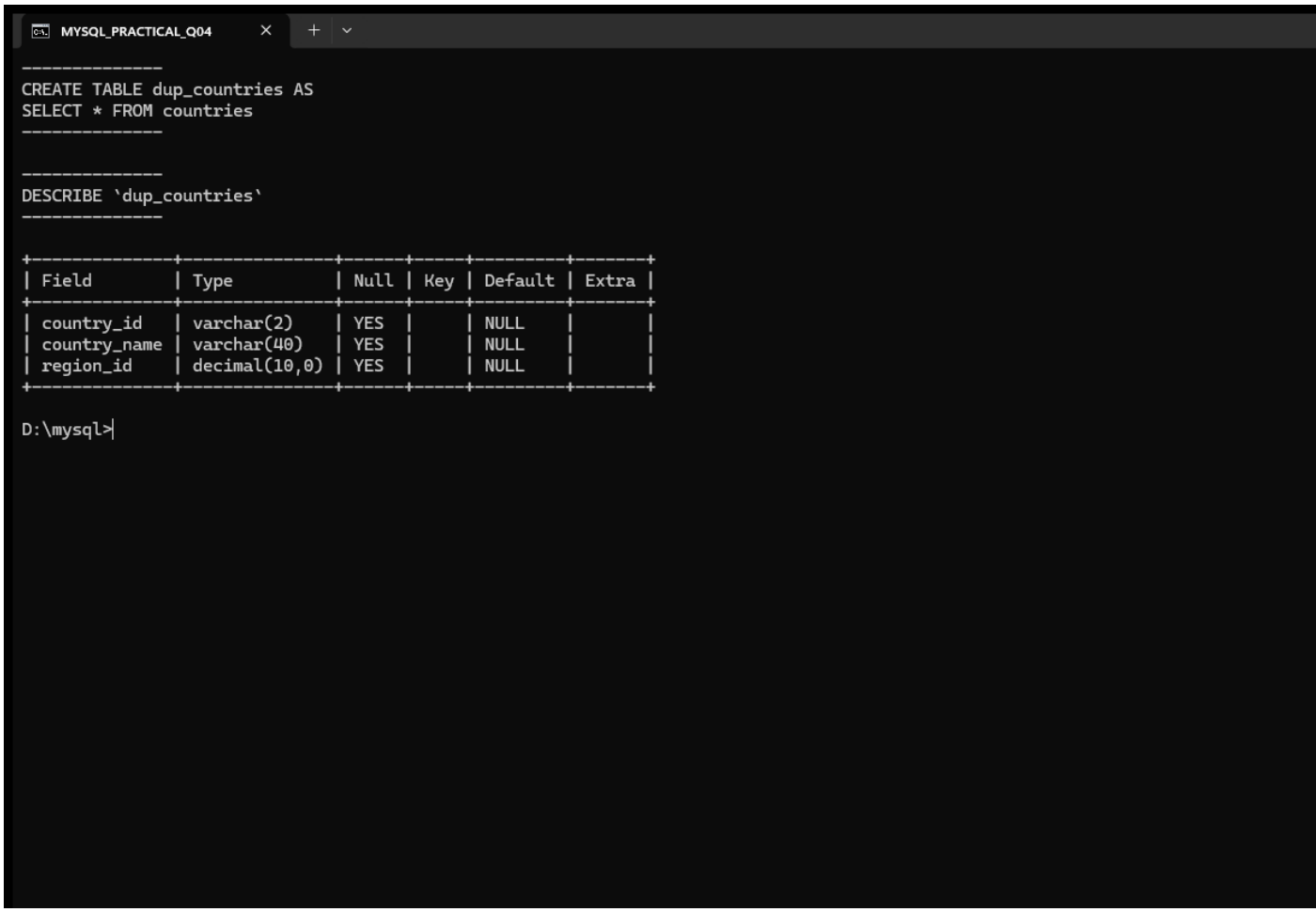
```
--------------
CREATE TABLE IF NOT EXISTS countries (
    country_id VARCHAR(2),
    country_name VARCHAR(40),
    region_id DECIMAL(10,0)
)
--------------

Note (Code 1050): Table 'countries' already exists
--------------
DESCRIBE `countries`
--------------

+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| country_id   | varchar(2)    | YES  |     | NULL    |       |
| country_name | varchar(40)   | YES  |     | NULL    |       |
| region_id    | decimal(10,0) | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+

D:\mysql>
```


MYSQL_PRACTICAL_Q03
CREATE TABLE dup_countries LIKE countries
DESCRIBE `dup_countries`
Field	Type	Null	Key	Default	Extra
country_id	varchar(2)	YES		NULL	
country_name	varchar(40)	YES		NULL	
region_id	decimal(10,0)	YES		NULL	
D:\mysql>


MYSQL_PRACTICAL_Q04
CREATE TABLE dup_countries AS
SELECT * FROM countries
DESCRIBE `dup_countries`
Field	Type	Null	Key	Default	Extra
country_id	varchar(2)	YES		NULL	
country_name	varchar(40)	YES		NULL	
region_id	decimal(10,0)	YES		NULL	
D:\mysql>

MYSQL_PRACTICAL_Q05

```
--------------
CREATE TABLE countries (
    country_id VARCHAR(2) NULL,
    country_name VARCHAR(40) NULL,
    region_id DECIMAL(10,0) NULL
)
--------------

--------------
DESCRIBE `countries`
--------------

+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| country_id   | varchar(2)    | YES  |     | NULL    |       |
| country_name | varchar(40)   | YES  |     | NULL    |       |
| region_id    | decimal(10,0) | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+

D:\mysql>
```

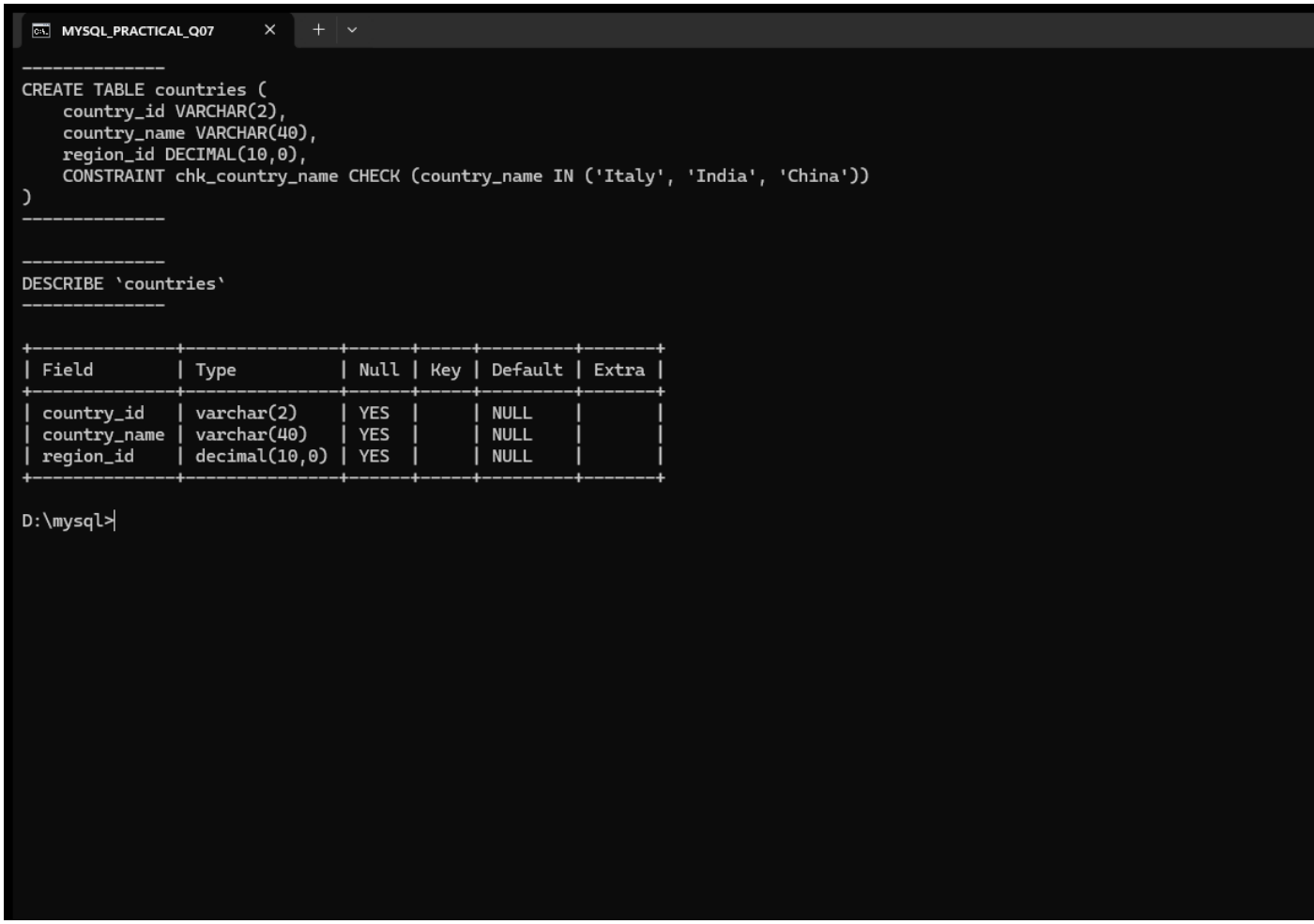
MYSQL_PRACTICAL_Q06

```
--------------
CREATE TABLE jobs (
    job_id INT,
    job_title VARCHAR(35),
    min_salary DECIMAL(6,0),
    max_salary DECIMAL(6,0),
    CONSTRAINT chk_max_salary CHECK (max_salary <= 25000)
)
--------------

--------------
DESCRIBE `jobs`
--------------

+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| job_id     | int          | YES  |     | NULL    |       |
| job_title  | varchar(35)  | YES  |     | NULL    |       |
| min_salary | decimal(6,0) | YES  |     | NULL    |       |
| max_salary | decimal(6,0) | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+

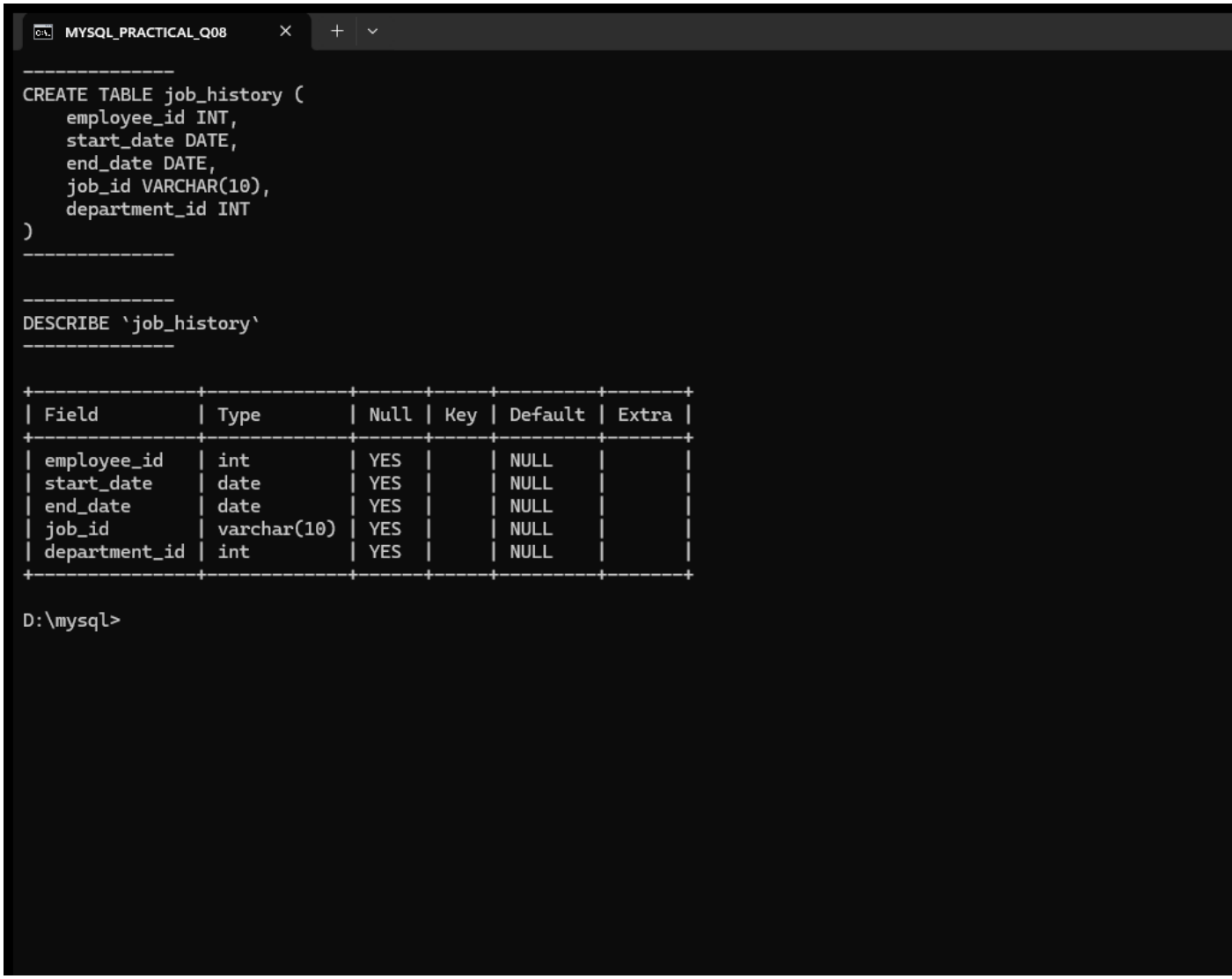
D:\mysql>
```

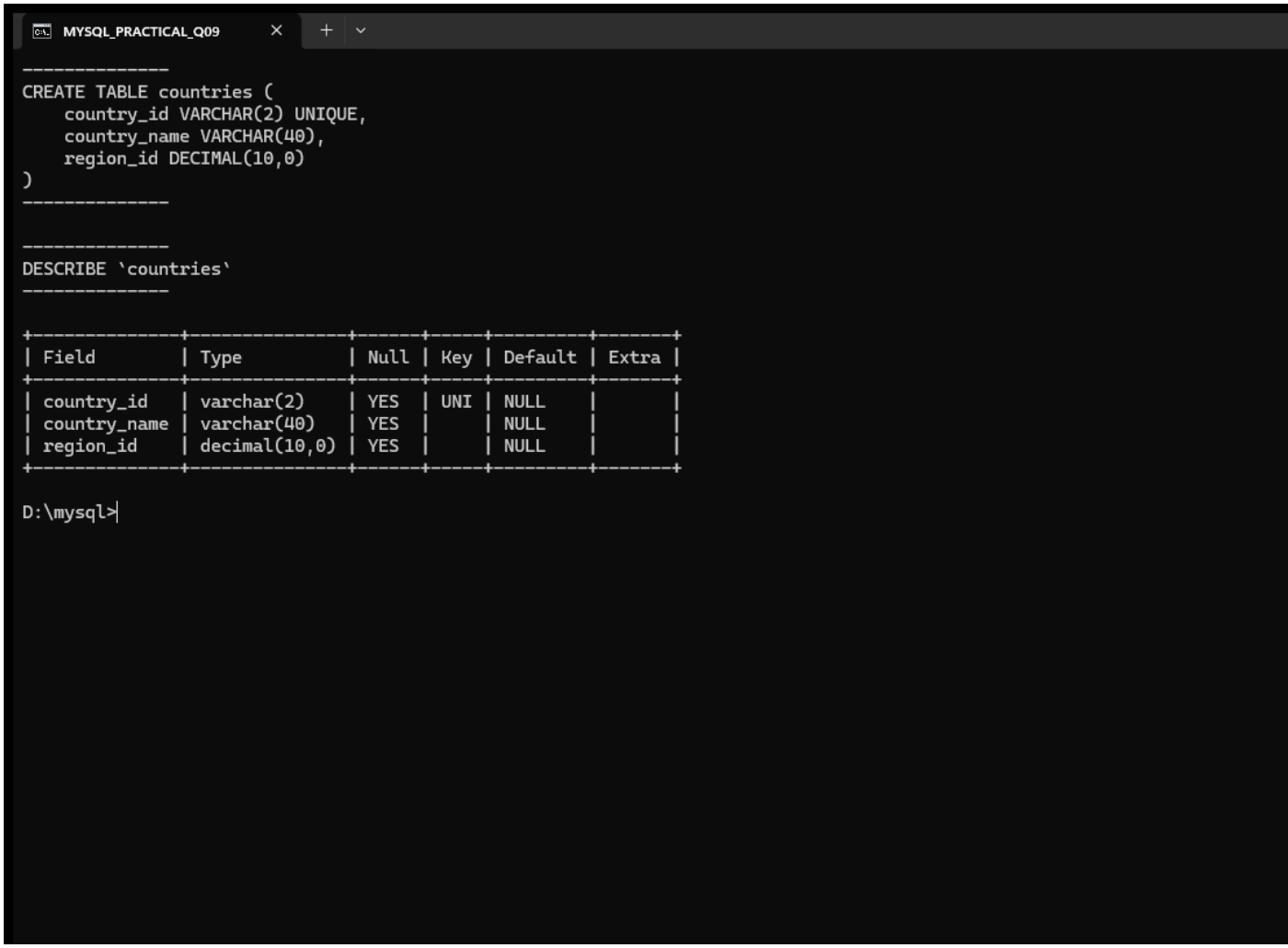

MYSQL_PRACTICAL_Q07

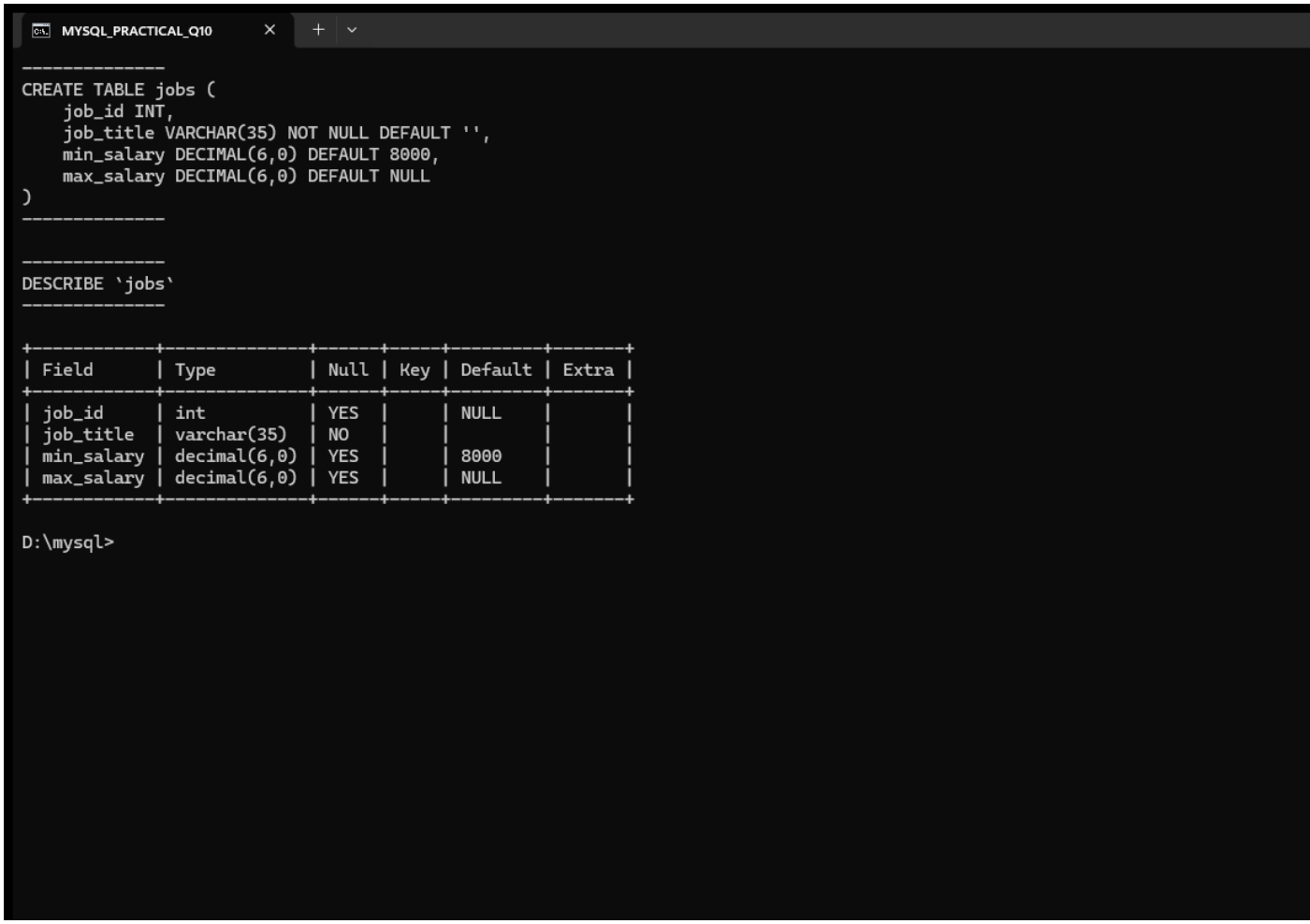
CREATE TABLE countries (
country_id VARCHAR(2),
country_name VARCHAR(40),
region_id DECIMAL(10,0),
CONSTRAINT chk_country_name CHECK (country_name IN ('Italy', 'India', 'China'))
)

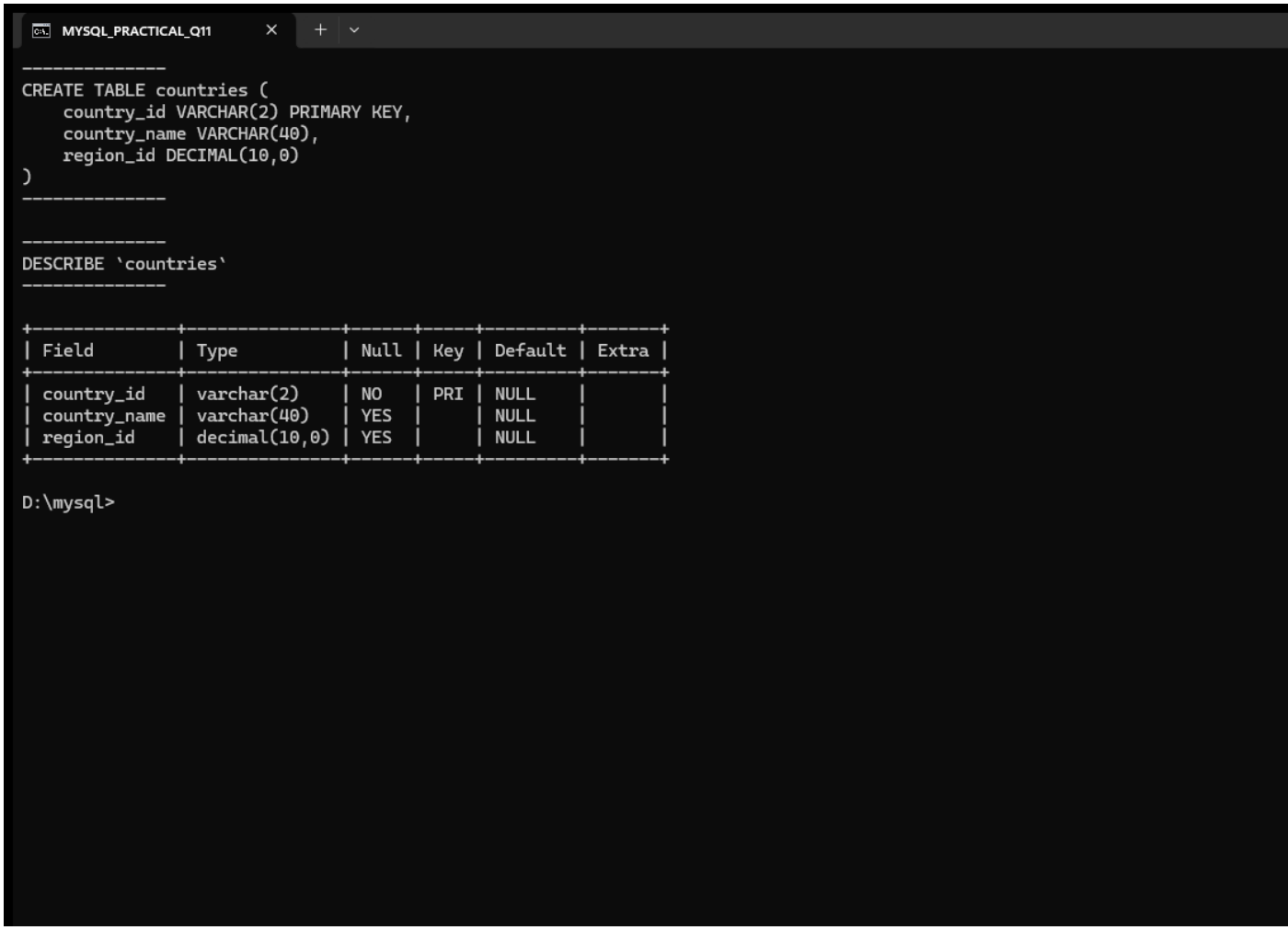
DESCRIBE `countries`

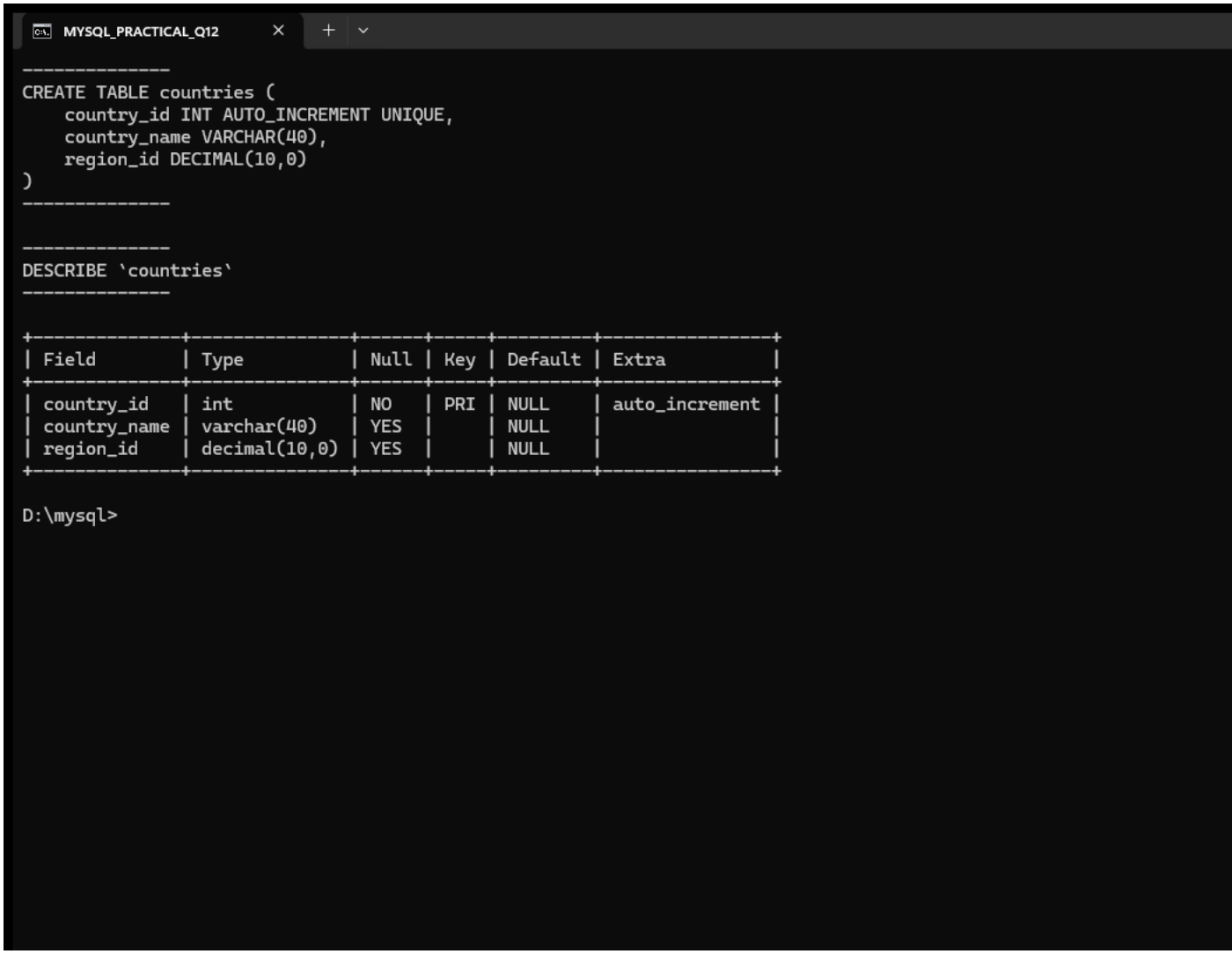
+--------------+---------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
country_id	varchar(2)	YES		NULL	
country_name	varchar(40)	YES		NULL	
region_id	decimal(10,0)	YES		NULL	
+--------------+---------------+------+-----+---------+-------+
D:\mysql>

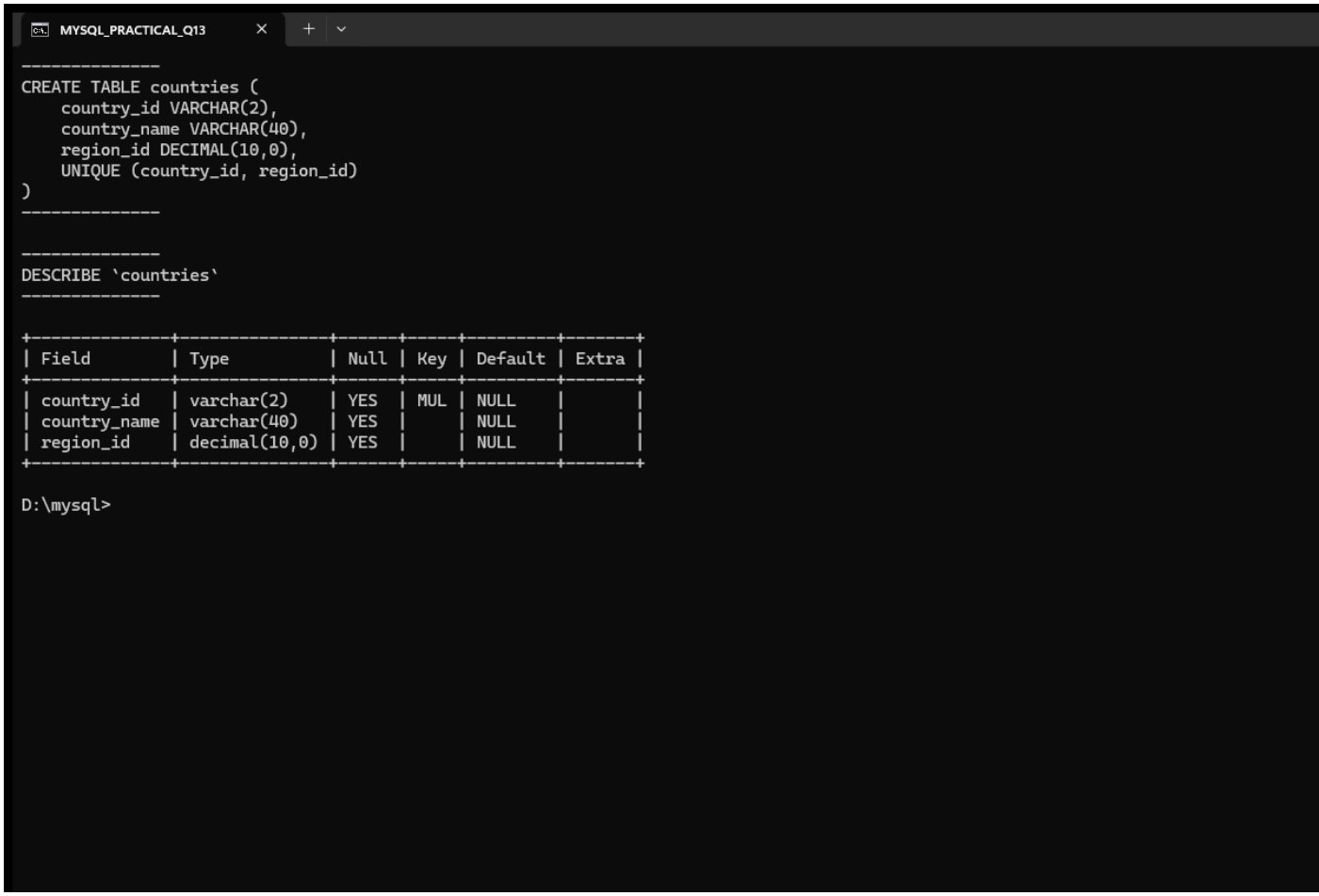

MYSQL_PRACTICAL_Q08
CREATE TABLE job_history (
employee_id INT,
start_date DATE,
end_date DATE,
job_id VARCHAR(10),
department_id INT
)
DESCRIBE `job_history`
Field	Type	Null	Key	Default	Extra
employee_id	int	YES		NULL	
start_date	date	YES		NULL	
end_date	date	YES		NULL	
job_id	varchar(10)	YES		NULL	
department_id	int	YES		NULL	
D:\mysql>

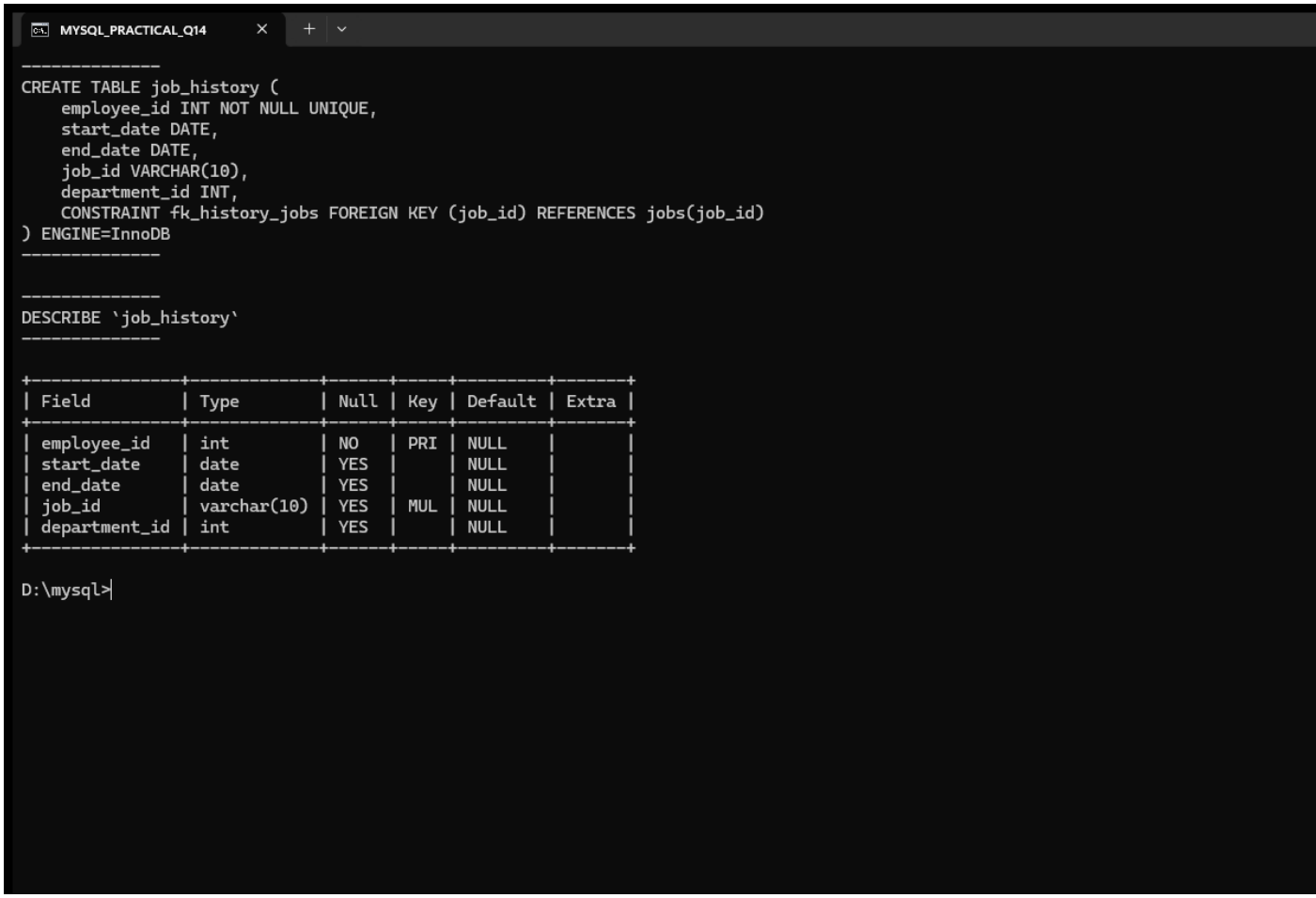

MYSQL_PRACTICAL_Q09
CREATE TABLE countries (
country_id VARCHAR(2) UNIQUE,
country_name VARCHAR(40),
region_id DECIMAL(10,0)
)
DESCRIBE `countries`
+--------------+---------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
country_id	varchar(2)	YES	UNI	NULL	
country_name	varchar(40)	YES		NULL	
region_id	decimal(10,0)	YES		NULL	
+--------------+---------------+------+-----+---------+-------+
D:\mysql>


MYSQL_PRACTICAL_Q10
CREATE TABLE jobs (
job_id INT,
job_title VARCHAR(35) NOT NULL DEFAULT '',
min_salary DECIMAL(6,0) DEFAULT 8000,
max_salary DECIMAL(6,0) DEFAULT NULL
)
DESCRIBE `jobs`
Field	Type	Null	Key	Default	Extra
job_id	int	YES		NULL	
job_title	varchar(35)	NO			
min_salary	decimal(6,0)	YES		8000	
max_salary	decimal(6,0)	YES		NULL	
D:\mysql>


MYSQL_PRACTICAL_Q11
CREATE TABLE countries (
country_id VARCHAR(2) PRIMARY KEY,
country_name VARCHAR(40),
region_id DECIMAL(10,0)
)
DESCRIBE `countries`
Field	Type	Null	Key	Default	Extra
country_id	varchar(2)	NO	PRI	NULL	
country_name	varchar(40)	YES		NULL	
region_id	decimal(10,0)	YES		NULL	
D:\mysql>


MYSQL_PRACTICAL_Q12
CREATE TABLE countries (
country_id INT AUTO_INCREMENT UNIQUE,
country_name VARCHAR(40),
region_id DECIMAL(10,0)
)
DESCRIBE `countries`
Field	Type	Null	Key	Default	Extra
country_id	int	NO	PRI	NULL	auto_increment
country_name	varchar(40)	YES		NULL	
region_id	decimal(10,0)	YES		NULL	
D:\mysql>


MYSQL_PRACTICAL_Q13
CREATE TABLE countries (
country_id VARCHAR(2),
country_name VARCHAR(40),
region_id DECIMAL(10,0),
UNIQUE (country_id, region_id)
)
DESCRIBE `countries`
Field	Type	Null	Key	Default	Extra
country_id	varchar(2)	YES	MUL	NULL	
country_name	varchar(40)	YES		NULL	
region_id	decimal(10,0)	YES		NULL	
D:\mysql>




```
mysql> CREATE TABLE employees (
    ->     employee_id INT NOT NULL PRIMARY KEY,
    ->     first_name VARCHAR(20),
    ->     last_name VARCHAR(25),
    ->     email VARCHAR(25),
    ->     phone_number VARCHAR(20),
    ->     hire_date DATE,
    ->     job_id VARCHAR(10),
    ->     salary DECIMAL(8,2),
    ->     commission DECIMAL(4,2),
    ->     manager_id DECIMAL(6,0),
    ->     department_id DECIMAL(4,0),
    ->     CONSTRAINT fk_employee_department_manager FOREIGN KEY (department_id, manager_id) REFERENCES departments(department_id, manager_id)
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected (0.04 sec)

mysql> desc employees;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| employee_id   | int         | NO   | PRI | NULL    |       |
| first_name    | varchar(20) | YES  |     | NULL    |       |
| last_name     | varchar(25) | YES  |     | NULL    |       |
| email         | varchar(25) | YES  |     | NULL    |       |
| phone_number  | varchar(20) | YES  |     | NULL    |       |
| hire_date     | date        | YES  |     | NULL    |       |
| job_id        | varchar(10) | YES  |     | NULL    |       |
| salary        | decimal(8,2)| YES  |     | NULL    |       |
| commission    | decimal(4,2)| YES  |     | NULL    |       |
| manager_id    | decimal(6,0)| YES  |     | NULL    |       |
| department_id | decimal(4,0)| YES  | MUL | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
11 rows in set (0.00 sec)

mysql>
```

```
mysql> CREATE TABLE employees (
    ->     employee_id INT NOT NULL PRIMARY KEY,
    ->     first_name VARCHAR(20),
    ->     last_name VARCHAR(25),
    ->     email VARCHAR(25),
    ->     phone_number VARCHAR(20),
    ->     hire_date DATE,
    ->     job_id VARCHAR(10),
    ->     salary DECIMAL(8,2),
    ->     commission DECIMAL(4,2),
    ->     manager_id DECIMAL(6,0),
    ->     department_id DECIMAL(4,0),
    ->     CONSTRAINT fk_emp_department FOREIGN KEY (department_id) REFERENCES departments(department_id),
    ->     CONSTRAINT fk_emp_job FOREIGN KEY (job_id) REFERENCES jobs(job_id)
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected (0.05 sec)

mysql> desc employees;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| employee_id   | int         | NO   | PRI | NULL    |       |
| first_name    | varchar(20) | YES  |     | NULL    |       |
| last_name     | varchar(25) | YES  |     | NULL    |       |
| email         | varchar(25) | YES  |     | NULL    |       |
| phone_number  | varchar(20) | YES  |     | NULL    |       |
| hire_date     | date        | YES  |     | NULL    |       |
| job_id        | varchar(10) | YES  | MUL | NULL    |       |
| salary        | decimal(8,2)| YES  |     | NULL    |       |
| commission    | decimal(4,2)| YES  |     | NULL    |       |
| manager_id    | decimal(6,0)| YES  |     | NULL    |       |
| department_id | decimal(4,0)| YES  | MUL | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
11 rows in set (0.00 sec)
```

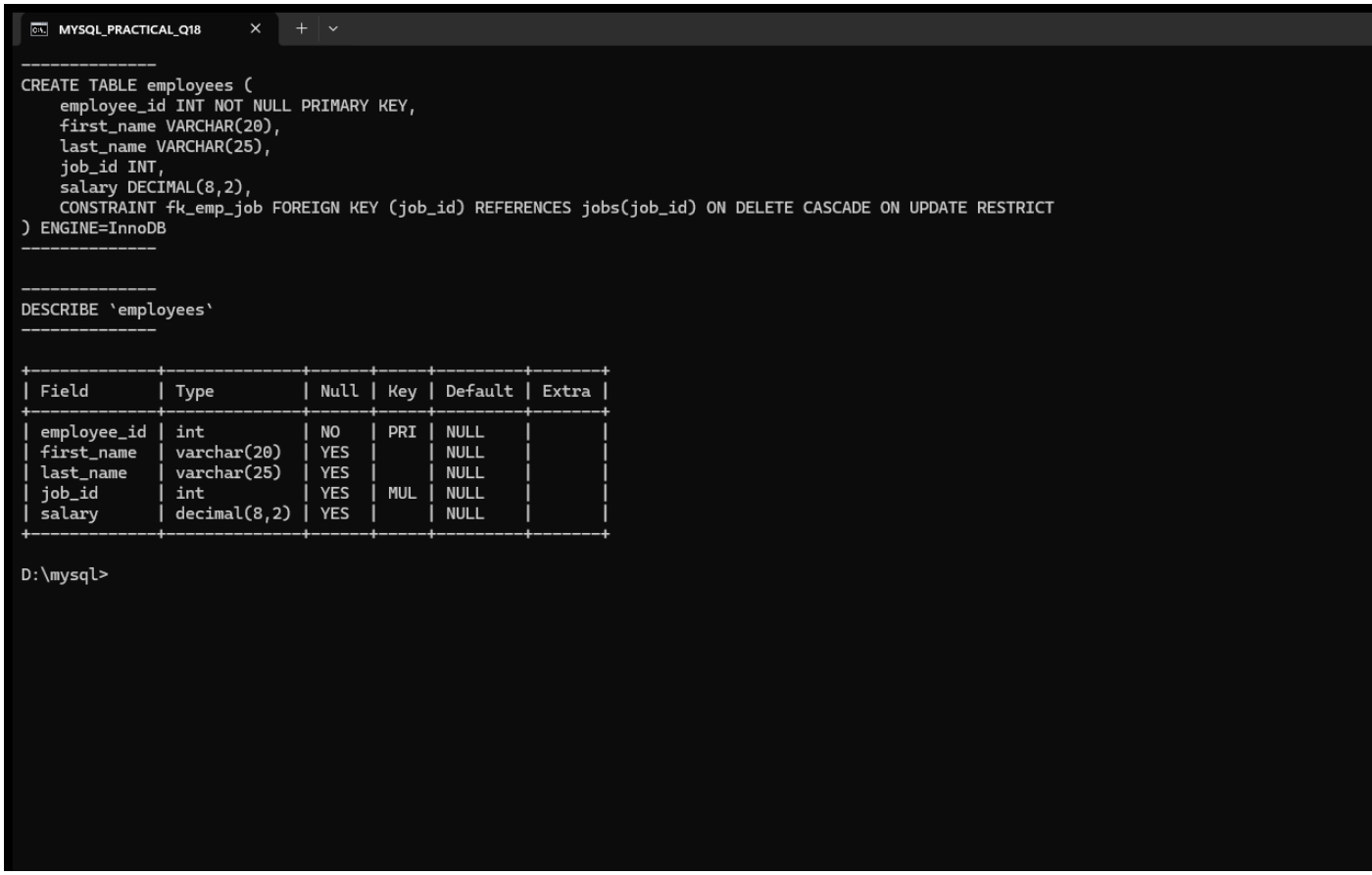
```
MYSQL_PRACTICAL_Q17

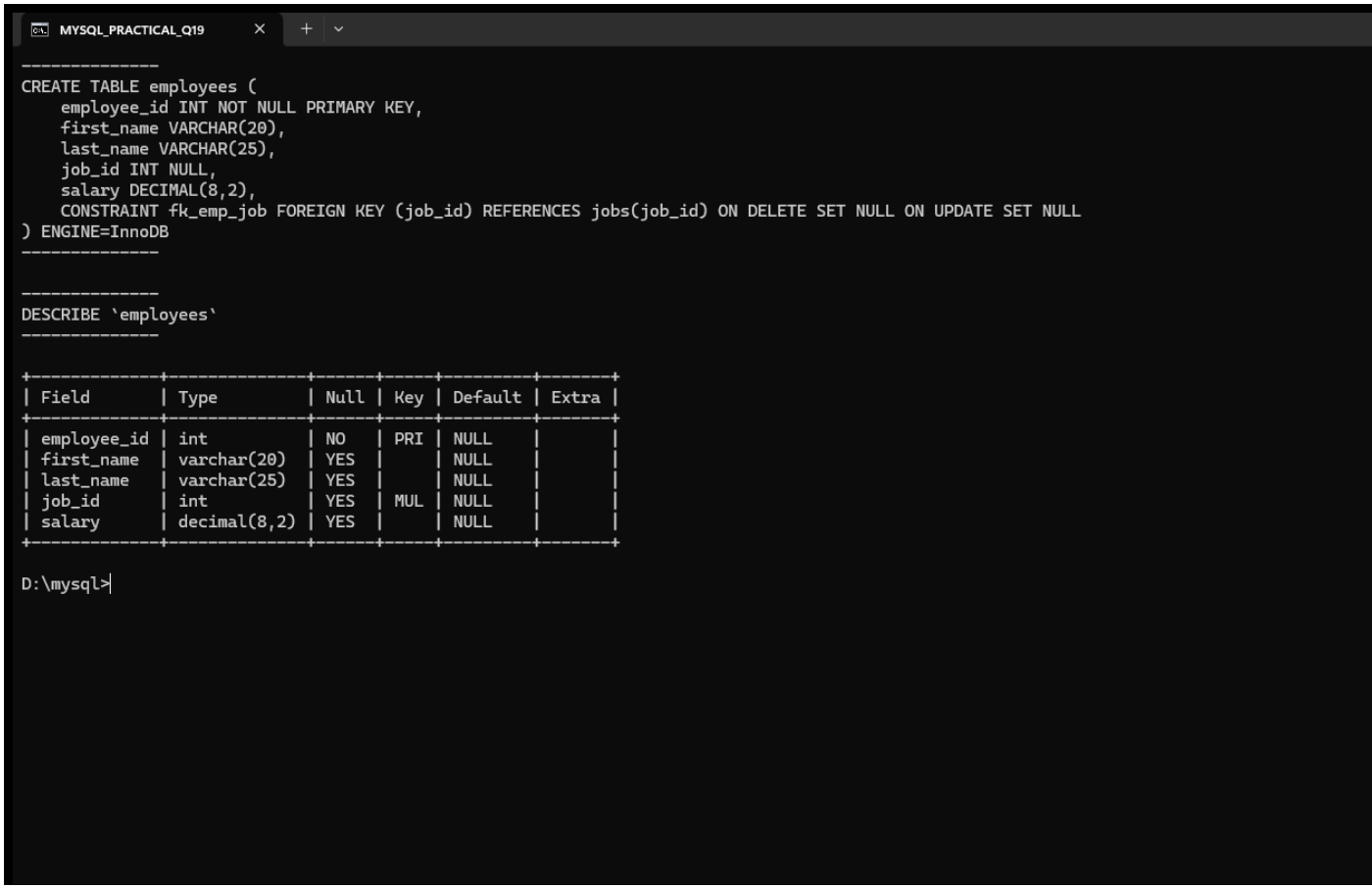
--------------
CREATE TABLE employees (
    employee_id INT NOT NULL PRIMARY KEY,
    first_name VARCHAR(20),
    last_name VARCHAR(25),
    job_id INT,
    salary DECIMAL(8,2),
    CONSTRAINT fk_emp_job FOREIGN KEY (job_id) REFERENCES jobs(job_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB
--------------

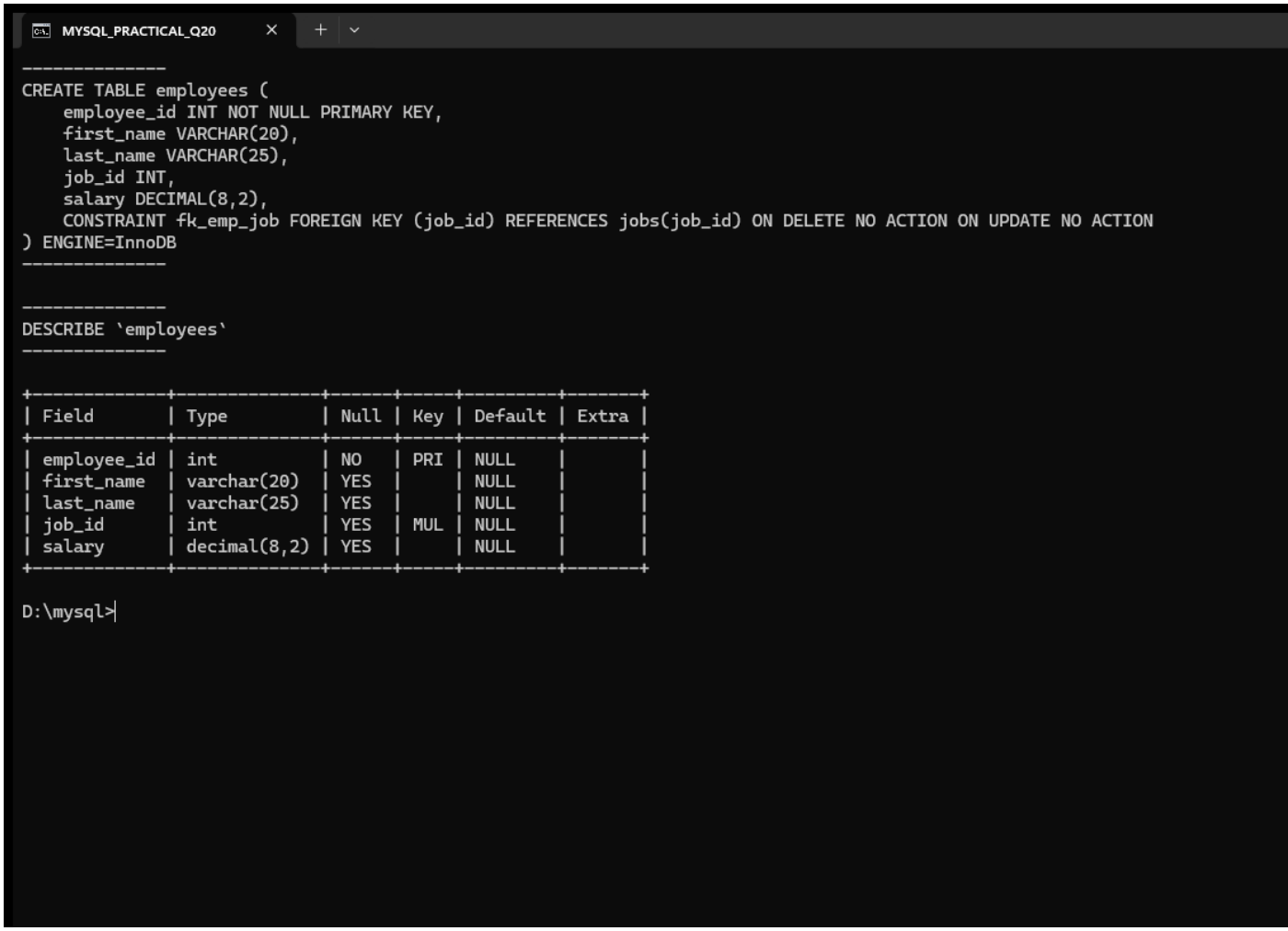
--------------
DESCRIBE `employees`
--------------

+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| employee_id | int          | NO   | PRI | NULL    |       |
| first_name  | varchar(20)  | YES  |     | NULL    |       |
| last_name   | varchar(25)  | YES  |     | NULL    |       |
| job_id      | int          | YES  | MUL | NULL    |       |
| salary      | decimal(8,2) | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+

D:\mysql>
```


MYSQL_PRACTICAL_Q18
CREATE TABLE employees (
employee_id INT NOT NULL PRIMARY KEY,
first_name VARCHAR(20),
last_name VARCHAR(25),
job_id INT,
salary DECIMAL(8,2),
CONSTRAINT fk_emp_job FOREIGN KEY (job_id) REFERENCES jobs(job_id) ON DELETE CASCADE ON UPDATE RESTRICT
) ENGINE=InnoDB
DESCRIBE `employees`
Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
first_name	varchar(20)	YES		NULL	
last_name	varchar(25)	YES		NULL	
job_id	int	YES	MUL	NULL	
salary	decimal(8,2)	YES		NULL	
D:\mysql>


MYSQL_PRACTICAL_Q19
CREATE TABLE employees (
employee_id INT NOT NULL PRIMARY KEY,
first_name VARCHAR(20),
last_name VARCHAR(25),
job_id INT NULL,
salary DECIMAL(8,2),
CONSTRAINT fk_emp_job FOREIGN KEY (job_id) REFERENCES jobs(job_id) ON DELETE SET NULL ON UPDATE SET NULL
) ENGINE=InnoDB
DESCRIBE `employees`
Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
first_name	varchar(20)	YES		NULL	
last_name	varchar(25)	YES		NULL	
job_id	int	YES	MUL	NULL	
salary	decimal(8,2)	YES		NULL	
D:\mysql>


MYSQL_PRACTICAL_Q20
CREATE TABLE employees (
employee_id INT NOT NULL PRIMARY KEY,
first_name VARCHAR(20),
last_name VARCHAR(25),
job_id INT,
salary DECIMAL(8,2),
CONSTRAINT fk_emp_job FOREIGN KEY (job_id) REFERENCES jobs(job_id) ON DELETE NO ACTION ON UPDATE NO ACTION
) ENGINE=InnoDB
DESCRIBE `employees`
Field Type Null Key Default Extra
employee_id int NO PRI NULL
first_name varchar(20) YES NULL
last_name varchar(25) YES NULL
job_id int YES MUL NULL
salary decimal(8,2) YES NULL
D:\mysql>